

Trader Behavior vs Market Sentiment

Data Science Assignment – Web3 Trading

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Introduction

The purpose of this project is to analyze how trader behavior changes under different market sentiment conditions in the cryptocurrency market. Specifically, this analysis studies the relationship between **Bitcoin market sentiment (Fear vs Greed)** and **trader performance and risk behavior** using historical trading data.

Understanding how traders behave during emotional market phases can help in designing smarter trading and risk management strategies.

Datasets Used:

1) Bitcoin Market Sentiment Dataset

This dataset provides daily market sentiment classifications:

- **Date**
- **Classification** (Fear / Greed)

It represents the overall emotional state of the Bitcoin market.

2) Historical Trader Data (Hyperliquid)

This dataset contains individual trade-level information, including:

- Account
- Trade size
- Trade side (Buy/Sell)

- Execution price
- Time
- Closed Profit and Loss (closedPnL)

Data Preparation & Methodology

- The **time** column in the trader dataset was converted into a date format.
- The sentiment dataset date was also standardized.
- Both datasets were merged using the **date** column so that each trade could be associated with the market sentiment on that day.
- Rows with missing critical values were removed to ensure data quality.

Note on Leverage:

Although leverage was mentioned in the assignment instructions, the provided trader dataset did not contain an explicit leverage column.

Therefore, **trade size and PnL volatility were used as proxy indicators** to evaluate trader risk-taking behavior.

Analysis & Key Findings

Profitability vs Market Sentiment

- Average trader profits were compared during Fear and Greed periods.
- Greed phases generally showed **higher profit potential**, but also increased variability in outcomes.

- Fear phases showed more cautious behavior but still contained large loss events.

Trade Size as a Risk Proxy

- Trade size was used to represent exposure and risk.
- Traders tended to place **larger trades during Greed periods**, indicating higher risk appetite.
- During Fear periods, trade sizes were relatively smaller.

PnL Volatility

- PnL volatility was higher during Greed phases.
- This indicates that traders experience larger gains and losses when market sentiment is optimistic.

Win Rate Analysis

- Win rate was calculated as the percentage of profitable trades.
- Win rates did not differ drastically between Fear and Greed, suggesting that risk behavior impacts magnitude of outcomes more than frequency of winning trades.

Loss Severity

- Losses during Fear periods tended to be more severe on average.
- This suggests panic-driven exits or unfavorable market conditions during fear phases.

Strategic Insights

Based on the analysis:

- Market sentiment strongly influences trader risk behavior.
- Greed phases encourage higher exposure and risk-taking.
- Fear phases can amplify loss severity.
- Market sentiment can be used as a **risk filter** rather than a direct trading signal.

Conclusion

This analysis demonstrates that trader behavior is closely linked to market sentiment. Emotional market phases influence risk appetite, trade sizing, and outcome variability. Incorporating sentiment awareness into trading strategies can improve risk-adjusted decision-making and capital protection.